

MINI PROJECT

1) Given Table

Member Id	First Name	Last Name	Hobbies
101	Jayson	Mark	Cricket, Swimming, Football
102	Ram	Ganesh	Swimming, Running, Music
103	Raj	Kishore	Dancing, Singing, Running

solution

Member Id	First Name	Last Name	Hobby
101	Jayson	Mark	Cricket
101	Jayson	Mark	Swimming
101	Jayson	Mark	Football
102	Ram	Ganesh	Swimming
102	Ram	Ganesh	Running
102	Ram	Ganesh	Music
103	Raj	Kishore	Dancing
103	Raj	Kishore	Singing
103	Raj	Kishore	Running

2) Empno Training Training Date Dept

101	Oracle SQL	12-Aug-2015	TT
101	Java	21-Aug-2015	BU
102	Oracle SQL	18-Sep-2014	TT

Dept depends only on Training, not on the entire composite key.

This is a partial dependency, so the table is not in 2NF.

Employee Training Table

Empno Training Training Date

101	Oracle SQL	12-Aug-2015
101	Java	21-Aug-2015
102	Oracle SQL	18-Sep-2014

Training Department Table

Training Dept

Oracle SQL	TT
Java	BU

3) Member Id First Name Last Name Sports Fees

101	Rajesh	Chand	Cricket	100
102	Jayesh	Raj	Hockey	80
103	Mark	Dorson	Football	90

This is a transitive dependency.

A table with transitive dependency is not in 3NF.

Member Table

Member Id	First Name	Last Name	Sports
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101	Rajesh	Chand	Cricket
102	Jayesh	Raj	Hockey
103	Mark	Dorson	Football

Sports Fees Table

Sports	Fees
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Cricket	100
Hockey	80
Football	90